

Nick Clouse

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SKILLS

Languages	Java, Python, JavaScript, R, HTML, CSS, SQL
Frameworks / Libraries / Tools	MySQL, PostgreSQL, React, Next.js, Spring/Spring Boot, Hibernate, JUnit, SciKit-Learn, NLTK, Flask, HoloViz, Git/GitHub/Gitlab

WORK EXPERIENCE

MSU - Space Science & Engineering Lab

May 2025 - August 2025

Software Engineering Intern

- Improved data parsing time by 62.5% by utilizing multiprocessing in Python.
- Developed a scalable data pipeline to process 1–2 GB of sensor data (measurements, health, position) from an instrument aboard the International Space Station in near real time.
- Built a Python-based web application with InfluxDB integration to visualize pipeline data, enabling easier monitoring and analysis.
- Engineered and optimized Ruby scripts to ensure fast, reliable data transfers between satellite and ground systems.

Iron Horse Golf Course

May 2023 - August 2024

Outside Services

- Provided high-quality member support by assisting guests and ensuring smooth operations during arrivals and events.
- Maintained and prepared golf equipment (clubs, carts, balls) with attention to detail, ensuring readiness and reliability.
- Supported tournament logistics by setting up equipment and coordinating event preparations.

PROJECTS

It Goes

Currently in development, *It Goes* is a full-stack web application for tracking and blogging ski trips, including Mapbox integration for backcountry route visualization. Built RESTful web services with Java & Spring (backend), connected to a Next.js & Tailwind CSS frontend, using PostgreSQL/PostGIS for spatial data and AWS S3 for image storage.

InitMusic

A full-stack web application to play music and build playlists. Built with a small team using Java, Spring, Gradle, JUnit, MySQL, HTML5, CSS, JavaScript, and RESTful web services. Integrated an external API for music streaming and utilized Git/GitHub for version control.

Avalanche Danger Classifier

Developed a Random Forest classifier that generates daily avalanche danger ratings for mountain ranges around Whitefish, Montana using HRRR weather data and SNOWPACK simulations. The model achieved $\sim 90\%$ train / $\sim 60\%$ test accuracy on 5 years of historical data, with ongoing work focused on improving generalization under anomalous snowpack conditions. Currently building a web interface to present daily danger ratings alongside an AI-generated weather and avalanche forecast.

EDUCATION

2022 - 2026 **B.A. Computer Science, B.A. Data Science** at Carroll College (Helena, MT) **GPA: 3.7/4.0**
Student-Athlete, starting on the football team as a kicker/punter

VOLUNTEERING

2026 FIRST Robotics Montana Qualifier